BloodLink

Blood bank inventory and compatible allocation, shipped through a full DevOps pipeline



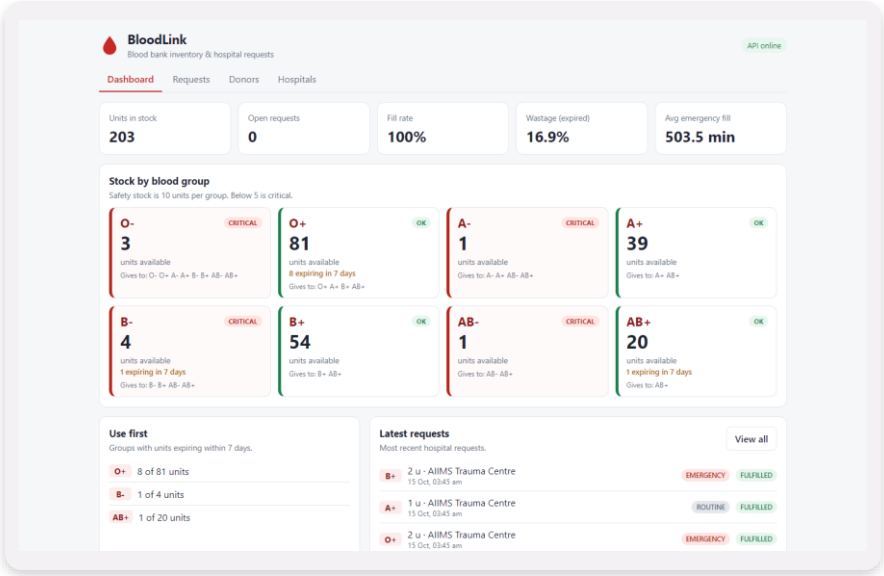
Ishan Ganguly · DevOps final capstone · github.com/Ishan007-bot/Bloodlink

Compatible blood, oldest unit first

1. **Compatible allocation**
   ABO/Rh rules: exact group first, then substitutes; O- kept for last

2. **First expiring, first out**
   Units expire 42 days after collection; the oldest usable unit is issued

3. **Emergency queue**
   Emergency requests are served before routine ones; partial fills are tracked



FastAPI · React · PostgreSQL 16 · 22 API routes · 5 tables managed by Alembic

From commit to running cluster



6 jobs

run on every push; images build only when tests and scans pass

1 SHA

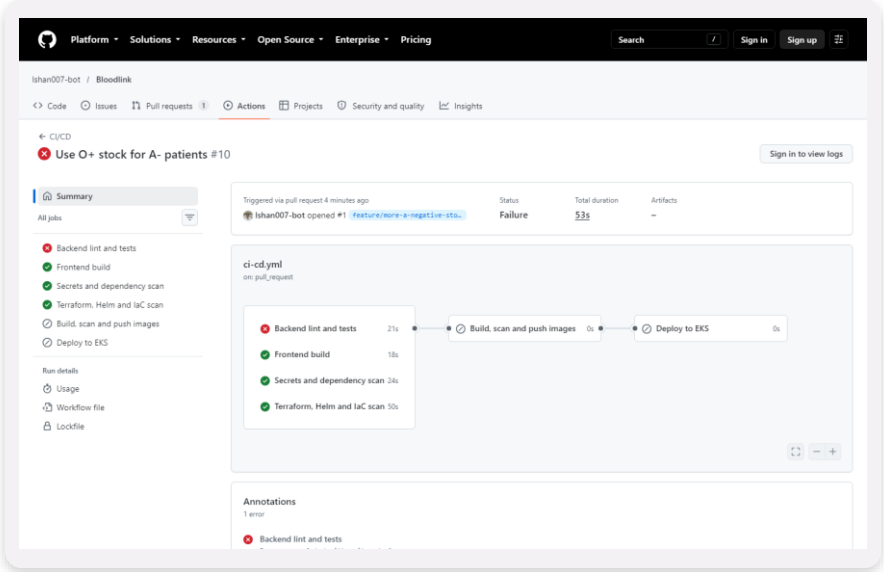
per image: every running pod traces back to one commit

0 keys

stored in GitHub: AWS access is short-lived OIDC

A failing test stopped an unsafe change

Pull request #1: “Use O+ stock for A- patients”
Looks helpful when A- stock is low, but gives Rh-positive blood to Rh-negative patients.



1 **Test fails**
is_compatible('O+', 'A-') returned True

2 **Nothing ships**
no image built, no deploy

3 **Fix, green, merge**
every check passes

81 tests · 97 % coverage · the build fails below 90 %

Security gates: 44 HIGH CVEs down to 0

44 → 0

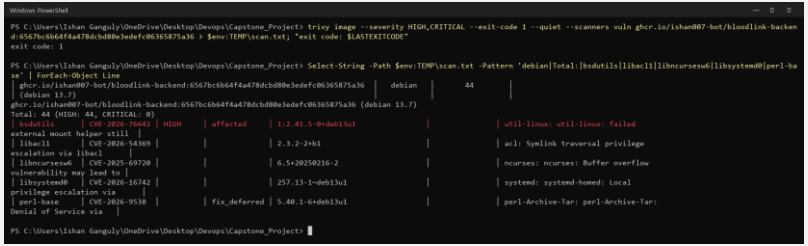
HIGH CVEs in the backend image

195 MB

image size, was 291 MB on Debian slim

One CVE explained: CVE-2026-76642 (util-linux)
mount runs privileged post-mount hooks even when the mount helper failed: local privilege escalation. Low risk here (non-root, read-only, no mount), but it shipped in every image, with no fix available. The gate now fails on unfixed CVEs too, and the backend moved to Alpine, which does not contain it.

- Trivy images + IaC
- gitleaks: no secrets
- pip-audit · npm audit
- non-root, read-only
- Pod Security restricted
- NetworkPolicy to Postgres
- OIDC, no AWS keys
- DB password in a Secret



AWS built and destroyed with Terraform

66

resources from one reviewed plan

2 + 2

public and private subnets in 2 zones

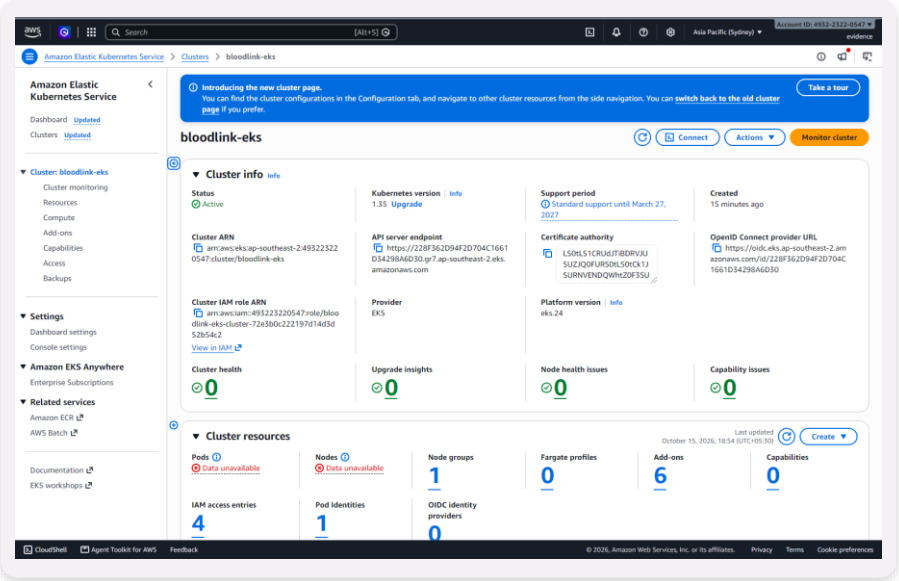
EKS 1.35

node group and six add-ons

≈ $0.50

one hour of running, then destroy

VPC with NAT gateway, KMS-encrypted secrets, S3 remote state, GitHub OIDC deploy role. Every apply and destroy ran from a saved plan. Region ap-southeast-2: the account's organization policy allows only that region.



terraform destroy: 66 destroyed, then nothing left to bill

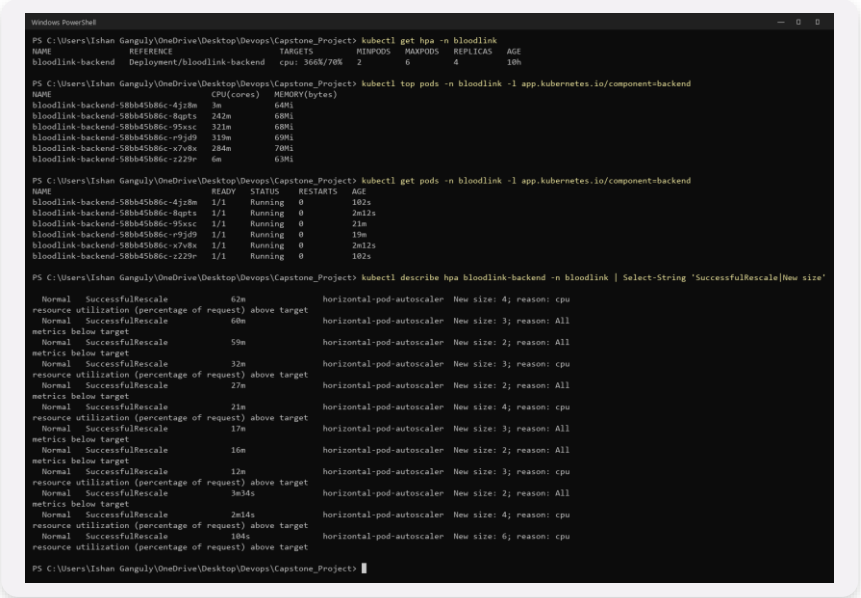
One Helm chart, from laptop to EKS

1. **Scales under load**
   HPA took the backend from 2 to 6 pods during a load test

2. **Migrations without races**
   init container + PostgreSQL advisory lock; rolling updates with no downtime

3. **Locked down**
   Pod Security restricted; only labelled clients reach Postgres

4. **Recoverable**
   nightly pg_dump CronJob, Helm rollback, 6 failure drills



Same chart: kind values locally, EKS values on AWS (gp3 volumes, Network Load Balancer)

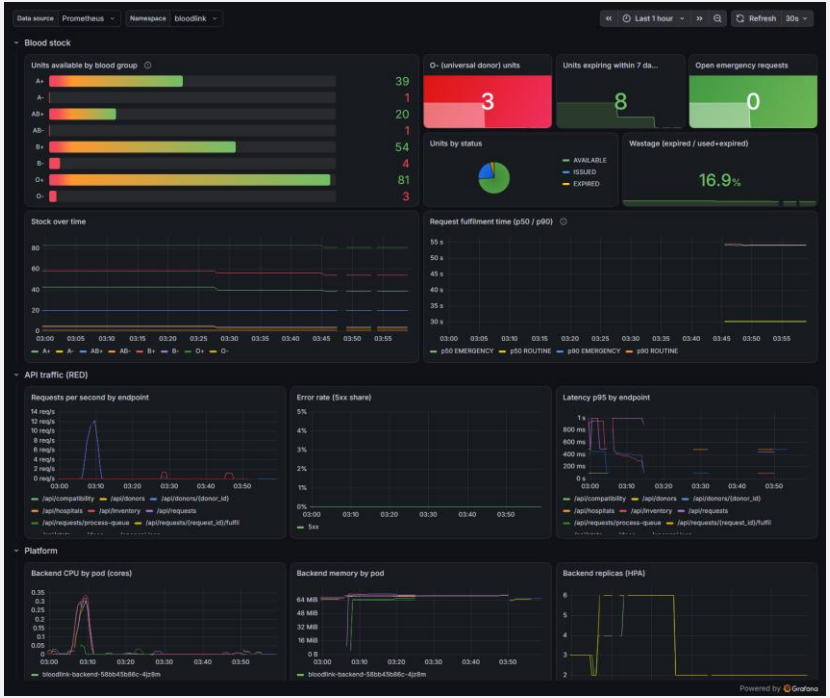
Metrics that matter to a blood bank

1. **Business metrics**
   stock per group, units expiring, fulfilment time

2. **API health**
   requests, error rate, p95 latency per route

3. **7 alert rules**
   critical stock and emergency backlog fire on real data

Prometheus scrapes every backend pod through a ServiceMonitor; the Grafana dashboard ships in the Helm chart.



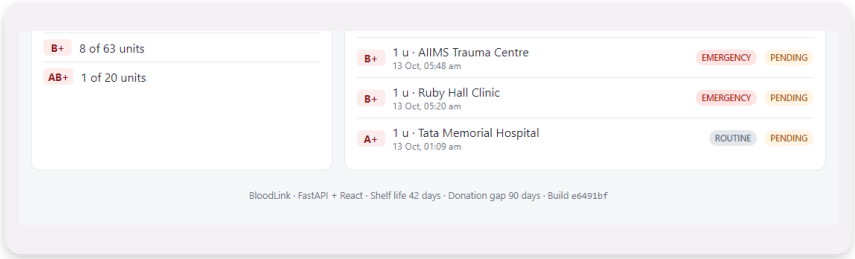
Live demo: commit → pipeline → deployment

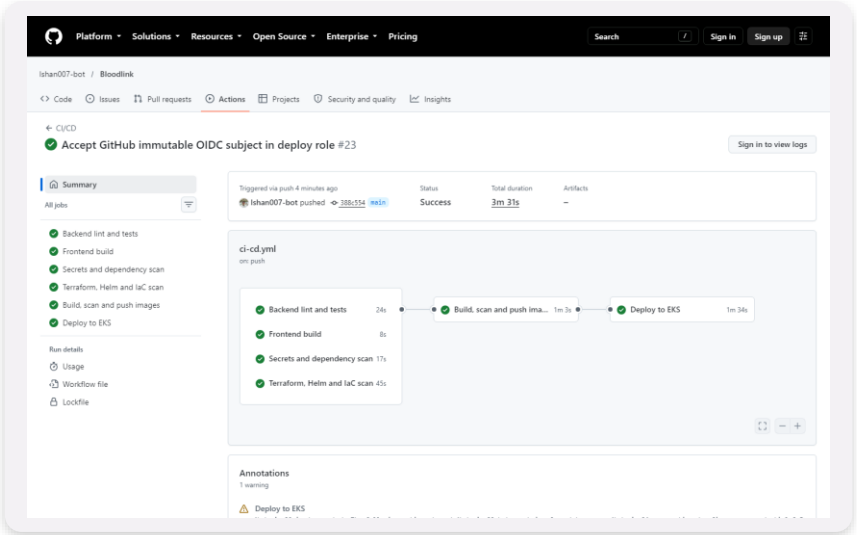
1 Commit
push a change to main

2 Pipeline
tests, scans, images: about 2 minutes

3 Deploy
helm upgrade to the new SHA

4 Verify
the footer shows the new build





Recorded run: docs/demo/bloodlink-demo.mp4 · on EKS the deploy job ran by itself through OIDC (commit 388c554)

What broke, and what it taught me

AWS allowed one region
dry-run checks per region showed only ap-southeast-2 passes the organization policy

GitHub could not sign in to AWS
CloudTrail showed the new immutable OIDC subject; fixed in the trust policy

Grafana kept restarting
OOMKilled at 256 MiB under load; limit raised to 512 MiB

Kubernetes control plane crashed
slow disk delayed etcd writes; leader election lost, tuned leases

Evidence for every module: docs/EVIDENCE.md · github.com/Ishan007-bot/Bloodlink